

## SUSSEX, NJ, USA

WMO: 740001

Lat: 41.199N

Lon: 74.626W

Elev: 125

StdP: 99.83

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54793

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -15.9      | -12.8 | -22.4                         | 0.5 | -13.7 | -19.6 | 0.7 | -10.8 | 7.9                   | -1.0 | 7.2 | -0.8 | 1.2                      | 220  | 0.440 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.8                   | 32.0            | 23.2 | 30.6 | 22.4 | 29.2 | 21.6 | 24.8                | 29.6 | 23.9 | 28.4 | 23.1 | 27.2 | 2.6                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.3                            | 18.4 | 27.2 | 22.5 | 17.5 | 26.1 | 21.9 | 16.8 | 25.3 | 75.6          | 29.5 | 72.0 | 28.5 | 69.0 | 27.3 | 27.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 7.0           | 5.7  | 5.1 | -20.3                      | 34.8 | 4.5                | 1.4 | -23.6                                              | 35.9 | -26.2      | 36.7 | -28.8      | 37.5 | -32.1      | 38.5 |    |    |
| (5)               |      |     | -20.6                      | 26.6 | 4.4                | 0.8 | -23.8                                              | 27.2 | -26.4      | 27.7 | -28.9      | 28.1 | -32.1      | 28.7 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.4   | -2.4  | -1.6  | 3.1   | 9.9   | 15.5  | 19.9  | 22.8  | 21.5  | 17.8  | 11.6  | 5.2   | 0.5  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.85   | 5.60  | 5.33  | 5.20  | 4.65  | 4.31  | 3.32  | 2.60  | 2.75  | 3.72  | 4.56  | 4.61  | 4.87 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1481   | 386   | 324   | 222   | 56    | 4     | 0     | 0     | 0     | 35    | 156   | 297   |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3282   | 644   | 557   | 472   | 254   | 110   | 22    | 2     | 6     | 55    | 213   | 394   | 554  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1622   | 1     | 1     | 8     | 54    | 176   | 298   | 396   | 356   | 234   | 86    | 12    | 2    | (10) |
| (11) |                                                                                  | CDD18.3     | 381    | 0     | 0     | 0     | 3     | 23    | 69    | 139   | 103   | 38    | 5     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 3528   | 0     | 0     | 2     | 66    | 291   | 660   | 1274  | 855   | 343   | 38    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1079   | 0     | 0     | 0     | 17    | 83    | 190   | 446   | 247   | 90    | 6     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.8    | 2.0   | 2.2   | 2.4   | 2.4   | 1.7   | 1.4   | 1.3   | 1.1   | 1.3   | 1.8   | 1.9   | 2.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1144   | 86    | 67    | 96    | 89    | 90    | 115   | 101   | 110   | 107   | 100   | 85    | 98   | (15) |
| (16) |                                                                                  | PrecMax     | 1707   | 197   | 181   | 220   | 194   | 198   | 249   | 186   | 329   | 288   | 369   | 130   | 200  | (16) |
| (17) |                                                                                  | PrecMin     | 686    | 40    | 17    | 21    | 43    | 22    | 24    | 27    | 33    | 2     | 0     | 32    | 29   | (17) |
| (18) |                                                                                  | PrecStd     | 242    | 39    | 32    | 47    | 38    | 40    | 59    | 39    | 61    | 79    | 82    | 28    | 43   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 16.0  | 17.2  | 22.5  | 29.2  | 31.3  | 32.9  | 34.3  | 32.8  | 31.7  | 27.0  | 21.0  | 16.4 | (19) |
| (20) |                                                                                  |             | MCWB   | 13.9  | 12.3  | 14.1  | 16.7  | 21.3  | 23.7  | 23.8  | 24.5  | 23.0  | 19.6  | 15.1  | 14.3 | (20) |
| (21) |                                                                                  | 2%          | DB     | 11.2  | 12.3  | 18.4  | 25.0  | 29.1  | 30.7  | 32.2  | 30.9  | 29.3  | 23.9  | 17.8  | 13.0 | (21) |
| (22) |                                                                                  |             | MCWB   | 8.7   | 8.4   | 11.5  | 15.2  | 20.1  | 22.3  | 23.6  | 23.2  | 21.6  | 18.3  | 13.7  | 10.8 | (22) |
| (23) |                                                                                  | 5%          | DB     | 7.4   | 8.7   | 14.3  | 22.1  | 26.8  | 28.8  | 30.7  | 29.2  | 27.1  | 21.9  | 15.5  | 10.0 | (23) |
| (24) |                                                                                  |             | MCWB   | 5.2   | 5.1   | 8.8   | 14.0  | 18.7  | 21.3  | 22.8  | 22.1  | 20.3  | 17.5  | 11.7  | 7.8  | (24) |
| (25) |                                                                                  | 10%         | DB     | 5.1   | 5.9   | 11.5  | 19.1  | 24.1  | 27.2  | 29.2  | 27.8  | 25.1  | 19.7  | 13.1  | 7.8  | (25) |
| (26) |                                                                                  |             | MCWB   | 2.9   | 2.9   | 7.4   | 11.9  | 17.2  | 20.3  | 22.1  | 21.4  | 19.6  | 15.6  | 10.0  | 5.4  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.2  | 14.1  | 15.4  | 19.2  | 23.0  | 24.7  | 25.8  | 26.2  | 24.8  | 22.1  | 17.3  | 15.0 | (27) |
| (28) |                                                                                  |             | MADB   | 15.4  | 16.8  | 19.7  | 24.5  | 29.1  | 30.5  | 31.4  | 31.2  | 29.3  | 24.6  | 19.7  | 16.3 | (28) |
| (29) |                                                                                  | 2%          | WB     | 8.6   | 9.0   | 12.6  | 17.2  | 21.5  | 23.7  | 24.8  | 24.7  | 23.2  | 20.3  | 14.8  | 11.2 | (29) |
| (30) |                                                                                  |             | MADB   | 10.7  | 11.9  | 17.3  | 23.8  | 27.2  | 28.7  | 29.9  | 28.8  | 26.8  | 22.7  | 17.1  | 12.6 | (30) |
| (31) |                                                                                  | 5%          | WB     | 5.5   | 5.5   | 10.0  | 15.0  | 20.2  | 22.7  | 24.0  | 23.5  | 21.9  | 18.5  | 12.6  | 8.0  | (31) |
| (32) |                                                                                  |             | MADB   | 7.2   | 8.2   | 13.4  | 20.2  | 24.7  | 27.3  | 28.6  | 27.3  | 25.4  | 21.2  | 14.3  | 9.6  | (32) |
| (33) |                                                                                  | 10%         | WB     | 3.2   | 3.1   | 7.7   | 12.9  | 18.8  | 21.5  | 23.2  | 22.5  | 20.8  | 16.3  | 10.3  | 5.6  | (33) |
| (34) |                                                                                  |             | MADB   | 4.7   | 5.4   | 11.3  | 17.3  | 22.6  | 25.4  | 27.4  | 26.0  | 23.5  | 18.9  | 12.8  | 7.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.4   | 10.3  | 10.7  | 12.6  | 12.6  | 11.9  | 11.8  | 11.7  | 12.2  | 11.5  | 10.5  | 8.5  | (35) |
| (36) |                                                                                  |             | MCDBR  | 12.0  | 14.1  | 15.3  | 17.7  | 16.0  | 14.1  | 13.7  | 13.0  | 14.4  | 14.0  | 13.9  | 11.0 | (36) |
| (37) |                                                                                  |             | MCWBR  | 10.0  | 10.5  | 9.4   | 9.1   | 8.0   | 6.8   | 6.2   | 6.1   | 7.4   | 8.4   | 9.9   | 9.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCDDBR | 11.5  | 13.0  | 13.3  | 14.6  | 13.1  | 12.4  | 11.3  | 11.1  | 11.2  | 11.3  | 11.5  | 10.1 | (38) |
| (39) |                                                                                  |             | MCWBR  | 10.2  | 10.3  | 9.4   | 8.9   | 7.4   | 6.6   | 5.6   | 5.7   | 6.4   | 7.9   | 9.8   | 9.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.300  | 0.320 | 0.350 | 0.370 | 0.414 | 0.475 | 0.479 | 0.463 | 0.382 | 0.356 | 0.321 | 0.301 | (40) |      |
| (41) |                                                                                  | taud        | 2.430  | 2.376 | 2.381 | 2.370 | 2.275 | 2.137 | 2.141 | 2.176 | 2.375 | 2.458 | 2.514 | 2.505 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 855    | 888   | 898   | 901   | 865   | 809   | 801   | 803   | 853   | 838   | 828   | 828   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 79     | 97    | 108   | 117   | 132   | 152   | 150   | 141   | 108   | 89    | 73    | 68    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.77   | 2.66  | 3.61  | 4.62  | 5.15  | 5.60  | 5.80  | 5.03  | 4.20  | 2.85  | 1.98  | 1.47  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.14   | 0.28  | 0.29  | 0.48  | 0.54  | 0.52  | 0.31  | 0.33  | 0.48  | 0.31  | 0.19  | 0.15  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (64 neighbors) | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +114    | +64     |

Nomenclature: See separate page